

① a) Black-Scholes formula for a European call.

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$\text{where } d_{1,2} = \frac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$C = 4.20$, $S_0 = 38$, $K = 35$, $r = 0.06$, $T = 4/12 = 0.3333$.
We try $\sigma = 0.1, 0.3, 0.5$.

$$\sigma = 0.1$$

$$T = \frac{1}{3}, \quad \ln(S_0/K) = \ln(38/35) = 0.819$$

$$d_1 = \frac{0.819 + (0.06 + 0.5 \times 0.01) \times \frac{1}{3}}{0.1 \sqrt{\frac{1}{3}}} = 1.023$$

$$d_2 = d_1 - 0.10 \sqrt{\frac{1}{3}} = 0.833$$

$$N(d_1) = 0.847, \quad N(d_2) = 0.797$$

$$\begin{aligned} C(0.10) &= 38(0.847) - 35e^{-0.06 \times \frac{1}{3}}(0.797) \\ &= 32.186 - 27.334 = 4.846 \end{aligned}$$

$$C(0.10) \approx 4.85 \approx 4.2$$

$$\sigma = 0.3$$

$$\sigma\sqrt{T} = 0.1732$$

$$d_1 = \frac{0.0819 + (0.06 + 0.5 \times 0.09) \times \frac{1}{3}}{0.1732} = 0.654$$

$$N(d_1) = 0.743, \quad N(d_2) = 0.685$$

$$\begin{aligned} C(0.3) &= 38(0.743) - 35e^{-0.06 \times \frac{1}{3}}(0.685) \\ &= 28.234 - 23.467 = 4.767 \end{aligned}$$

$$4.77 > 4.2$$

$$\sigma = 0.5$$

$$\sigma\sqrt{T} = 0.2887$$

$$d_1 = \frac{0.0819 + (0.06 + 0.5 \times 0.25) \times 0.333}{0.2887} = 0.402$$

$$d_2 = 0.402 - 0.2887 = 0.113$$

$$N(d_1) = 0.657, \quad N(d_2) = 0.545$$

$$\begin{aligned} C(0.5) &= 38(0.657) - 35(0.982)(0.545) \\ &= 6.32 \end{aligned}$$

$$6.32 > 4.2$$

By interpolation,

we see call price decreases between 0.1 and 0.3
if we take 0.25, so $\sigma = 0.25$.

$$b) \quad C - P = S_0 - Ke^{-rT}$$

$$\Rightarrow P = C - S_0 + Ke^{-rT}$$

$$C = 9.2, \quad S_0 = 38, \quad K = 35, \quad e^{-rT} = e^{-0.06 \times \frac{1}{3}} = 0.9802$$

$$P = 9.2 - 38 + 35(0.9802)$$

$$= 9.2 - 38 + 34.307 = 0.507.$$

\$0.51. $\rightarrow$ put price

a) launch opportunity as a call option -

$$S_0 = 38, \quad K = 35, \quad T = \frac{1}{3}, \quad r = 0.06, \quad \sigma = 0.28.$$

$$\sigma\sqrt{T} = 0.28 \sqrt{\frac{1}{3}} = 0.1617,$$

$$d_1 = \ln(38/35) + (0.06 + 0.5 \times 0.28^2) \frac{1}{3} = 0.693$$

$$d_2 = 0.693 - 0.1617 = 0.531$$

$$C = 38 N(0.693) - 35e^{-0.06 \times \frac{1}{3}} N(0.531)$$

$$= 38 (0.756) - 35 (0.9802) (0.703)$$

$$= 28.728 - 24.045 = \$4.683$$

$$S_0 - K = \$3$$

$C(0.28) > S_0 - K$, value of waiting exceeds NPV.

Hence, firm should delay the project.

Assignment-4

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(b)
Solution

Que-2]

Part-A: Discrete Binomial model

o S_0 (current stock price) = 100

o strike price $K = 105$

o Time to maturity $T = 10$ days

Q (a) probability that the option ends in money after 10 days

$$S_{10} > 105$$

$$2K + 90 > 105 \Rightarrow 2K > 15 \Rightarrow K > 7.5$$

~~that~~ $K \geq 8$

~~So valid~~ so $K = 8, 9, 10$

total no. of paths in binomial model; $2^{10} = 1024$

$$P(K) = \binom{10}{K} \cdot \left(\frac{1}{2}\right)^{10}$$

$$P(S_{10} > 105) = \sum_{K=8}^{10} \binom{10}{K} \cdot \left(\frac{1}{2}\right)^{10} = \frac{1}{1024} \left[\binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right]$$

$$= \frac{1}{1024} (45 + 10 + 1) = \frac{56}{1024}$$

$$\approx 0.05468 \quad \boxed{7/128} \quad \underline{\underline{\text{Ans}}}$$

(b) Solution $\rightarrow$ if $K = 8$

$$S = 2(8) + 90 = 106, \text{ Payoff} = 106 - 105 = 1$$

$$\text{Paths: } \binom{10}{8} = 45$$

if $K = 9$

$$S = 108, \text{ Payoff} = 3$$

$$\text{Paths: } \binom{10}{9} = 10$$

if $K = 10$

$$S = 110, \text{ Payoff} = 5$$

$$\text{Paths: } \binom{10}{10} = 1$$

total expected payoffs

$$E(\text{Payoff}) = \frac{1}{2024} [45 \times 1 + 10 \times 3 + 5 \times 5]$$

$$= \boxed{\frac{5}{64}} \text{ Ans}$$

(c) ~~as we~~ Ignoring Discounting

fair value = expected ~~value~~ payoff

$$= \boxed{5/64}$$

Part B

(a)

$$E[|x|] = \sigma \sqrt{\frac{2}{\pi}} = 1 \Rightarrow \sigma = \sqrt{\frac{\pi}{2}} \approx 1.2533$$

returns are additive and normal, over 10 days

$$\sigma_{10} = \sqrt{10} \cdot \sigma \approx 3.962$$

Daily standard deviation $\sigma \approx 1.2533$

10 day std deviation $\sigma_{10} \approx 3.962$

(b) Expected payoff

$$E[\max(S_T - K, 0)] = \int_K^\infty (S - K) f_{S_T}(S) dS$$

Let $\mu = 100$, $\sigma = 3.962$

$$f_{S_T}(S) = \frac{1}{\sqrt{2\pi} \cdot 3.962} \exp\left(-\frac{(S-100)^2}{2 \times 3.962^2}\right)$$

$$E[\max(S_T - 105, 0)] = \int_{105}^\infty (S - 105) \times f_{S_T}(S) dS$$

we will put the value of $f_{S_T}(S)$ from the above equation to get answer

integral numerically

$$c = (S_0 - K) \times \Phi\left(\frac{S_0 - K}{\sigma_T}\right) + \sigma_T \times \phi\left(\frac{S_0 - K}{\sigma_T}\right)$$

$$S_0 = 100, \quad K = 105, \quad \sigma_T = 3.962$$

$\Phi(z)$ is CDF of standard normal

$\phi(z)$ is PDF of standard normal

$$z = \frac{100 - 105}{3.962} = -1.261$$

$$\Phi(z) = \Phi(-1.261) \approx 0.1037$$

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \approx 0.1812$$

$$c = (100 - 105)(0.1037) + 3.962 \times 0.1812$$

$$= 0.1992$$

Part C

(a) Define support $[a, b]$ such that $E[|x|] = 1$

Let $x \sim \text{Uniform}[-x, x]$ for symmetric uniform distribution

$$E[|x|] = \int_{-x}^x \frac{|t|}{2x} dt = \frac{1}{x} \int_0^x t dt = \frac{1}{x} \cdot \frac{x^2}{2} = \frac{x}{2} = 1 \Rightarrow x = 2$$

$$\text{Support} = [-2, 2]$$

(b) * Uniform model leads to broader range, more probability in tails.

* normal mode is centered, symmetric, less tail weight

* Binomial is discrete, symmetric, medium spread.

↳ Binomial model ends in values from 90 to 110 in steps of 2

↳ normal model is continuous, Bell shaped distribution centered at 100.

↳ uniform model Each day moves in $[-2, 2]$

so 10 day change in $[-20, 20]$. final price $[80, 120]$. approximately uniform, but more like triangular distribution due to convolution.